Title: Familiarity enhances coordination and efficiency in complex collective tasks

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Abstract:
Familiarity among group members is thought to enhance the performance of animal groups facing complex ecological challenges. In humans, stable group membership—often termed "team familiarity"—improves coordination and efficiency. To test whether this principle extends to non-human collectives, we examined natural groups of Spix's disc-winged bats (*Thyroptera tricolor*) and experimentally assembled unfamiliar groups. Using three-dimensional flight reconstructions and roost-entry trials, we show that familiar groups maintain closer spatial cohesion, enter roosts in tighter synchrony, and collectively produce more acoustic signals while still minimizing signal overlap. These findings confirm that familiarity facilitates communication and coordination, extending the concept of team familiarity beyond humans and highlighting a general principle of collective success in animal societies.

Main Text: Animals can express remarkable collective behaviours, such as ants linking their bodies to form living bridges (*1*), honeybee swarms reaching consensus on the location of a new nest site (*2*) or predators coordinating group hunts (*3*). Although these feats may seem complex, they often emerge from simple rules. For example, local interactions among individuals can allow groups to track complex environmental gradients (*4*) while effectively detecting and avoiding predators (*5*, *6*). One feature of these simple rule-based collective actions is that they can operate despite the individual constituents being largely anonymous, as is widely observed in

schools of fish (*7*) and flocks of birds (*8*). Yet, human groups with stable membership consistently outperform those with fluctuating membership (*9–11*), especially when decisions require greater cooperation and efficient communication (*12*, *13*). This phenomenon has been formalized as the concept of "team familiarity", which highlights how repeated experiences among the same individuals improve their coordination and performance (*14*, *15*). An important question is therefore whether familiarity among group members also enhances the ability of other animal collectives to solve challenges that demand particularly intricate coordination and communication.

We experimentally tested whether familiarity increases group performance during collective tasks by non-human animals. Laboratory experiments using fish have suggested that familiarity can increase coordination in schools (*16*, *17*) and facilitate transmission of information (*18*, *19*). However, testing this link in natural groups is difficult because high familiarity is intrinsically linked with strong spatial cohesion and living in stable groups, such as groups of primates (*20*) and birds (*21*) where group membership rarely varies. This limits our ability to compare performance of collectives at solving ecologically relevant challenges across both familiar and unfamiliar settings. We overcome this challenge by bringing wild-caught groups of Spix's disc-winged bat (*Thyroptera tricolor*), a Neotropical species characterized by high social stability, temporarily (< 6 h) into captivity. This species forms long-term stable groups of about five individuals (range 2–9) that sleep inside developing tubular leaves of plants in the order Zingiberales, such as heliconias and bananas (*22–24*). Because these leaves unfurl after approximately one day, groups must repeatedly find new sleeping sites (*25*). Group members also have to enter the leaf in rapid succession to reduce both individual predation risk and overall group detectability (*26*, *27*). This spatial and temporal ecological challenge is readily replicated in laboratory conditions.

Group flight coordination

We used videos from four cameras to reconstruct the 3-dimensional trajectories of individuals in natural (high familiarity) groups and in artificial groups (novel combinations of individuals from several natural groups) within a large flight cage (9 x 4 x 3 m; Fig. 1A-B). If familiarity plays an important role in collective behaviours, we expected that the high familiarity groups would show greater spatial cohesion than the artificial groups. Annotating 5 s segments where the trajectories of all individuals were visible confirmed this prediction, with the data revealing that bats in natural groups flew significantly closer to one-another than bats in artificial groups (Fig. 2). Artificial groups did not differ from random expectation ('simulated groups' where the flight trajectories of individuals from different trials were digitally combined; Fig. 1C and 2), confirming that familiarity facilitates greater coordination.

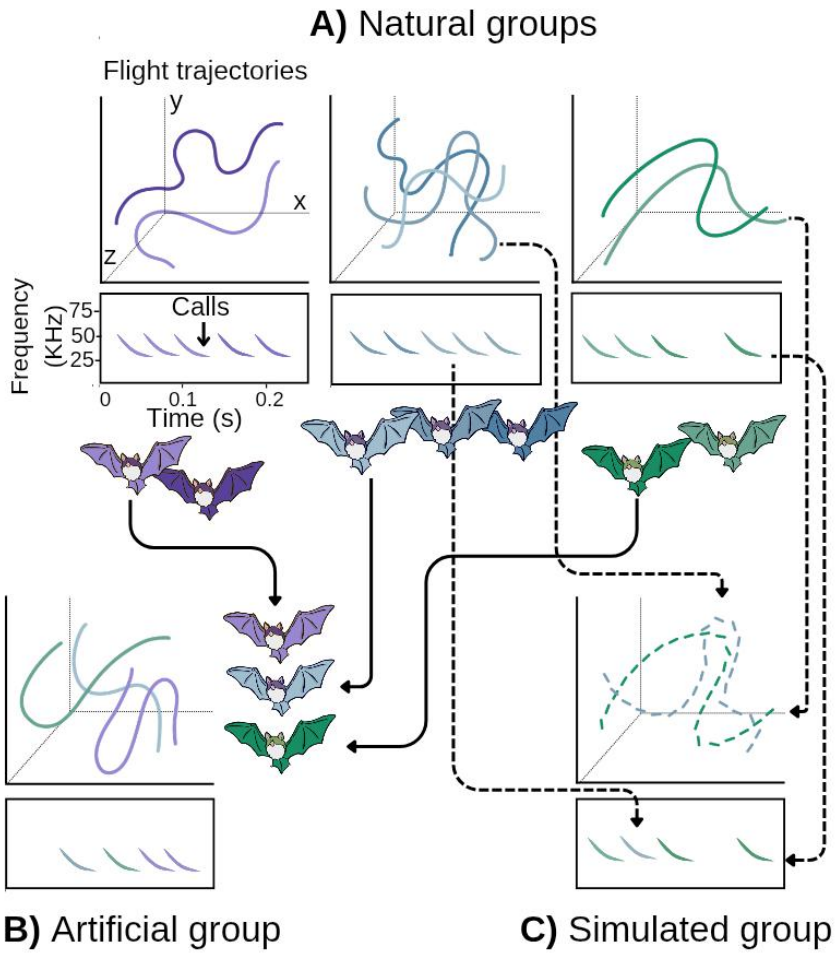

Fig. 1. Schematic of the experimental design used to assess group flight and calling coordination across different group conditions. Exemplary natural **(A)** and artificial **(B)** groups are shown next to a schematic 3-D flight trajectory and a spectrogram of the vocal activity with multiple inquiry calls. Natural group membership is depicted by color tonality and the trajectories and calls from a given individual match its color. Artificial groups were composed by temporarily combining individuals from different natural groups (depicted by solid line arrows from natural to artificial group bats) while simulated groups (**C**) were digitally created by combining flight trajectories and calling activities collected from individuals in natural group trials (illustrated by the dashed line arrows from the trajectories and calls from natural to simulated groups).

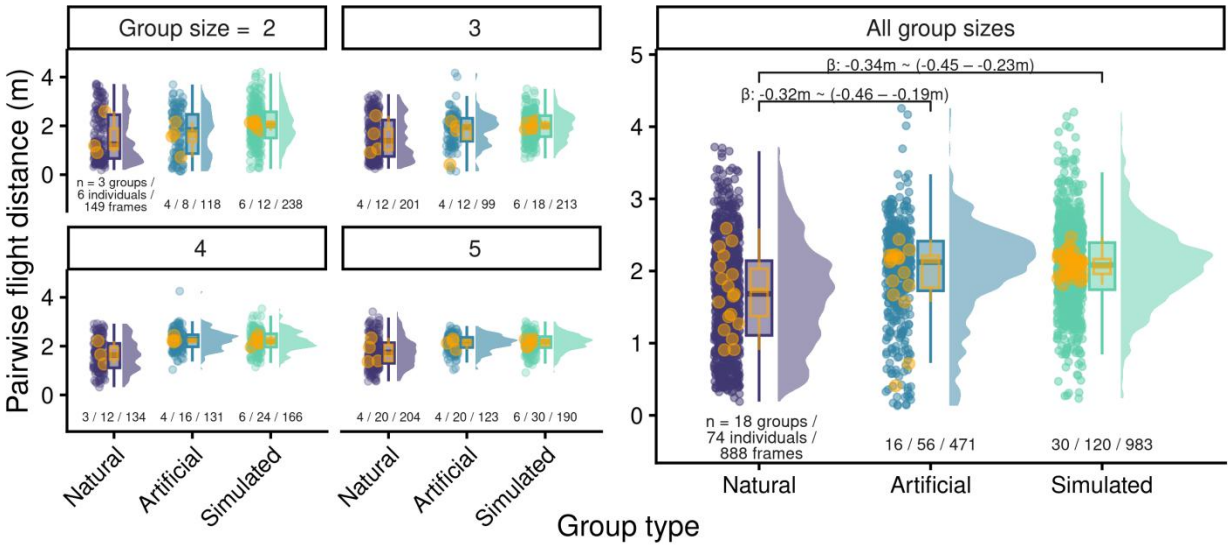

Fig. 2. Variation in pairwise flight distance between group types. Raincloud plot showing the jittered pairwise distance among individuals per video frame and the associated boxplots (median, interquartile range and overall range) and half violin plots, by group flight type (horizontal axis) and group size (panels). For simplicity, data for groups of six individuals are not shown. The right panel shows aggregated data for all group sizes. Yellow points show the mean pairwise distance per group with their distribution represented by overlaid yellow boxplots. Brackets above boxplots in the right panel indicate statistically detectable differences with effect size and 95% uncertainty intervals (in m). Sample sizes for groups, individuals and video frames are shown underneath each boxplot. Full model structure and results are available in the supplementary materials.

Roost entry synchrony

Synchronising actions plays an important role in reducing predation risk across a range of contexts. For example, breeding groups of birds often synchronise their provisioning by approaching the nest together to reduce nest visitation rates and thus avoiding detection of nest location by predators (*28*, *29*). Groups of Spix's disc-winged bats face this same challenge on a daily basis, as they require first finding a suitable new leaf in which to roost and then coordinating their entry to reduce the chances of detection by a range of avian and mammalian predators. Placing a furled leaf of *Heliconia* sp. or *Calathea lutea* at random positions (Fig. 3 A-C) within the flight cage and measuring the time from release into the flight cage until entering the leaf confirmed that natural (familiar) groups were substantially more synchronised than artificial (unfamiliar) groups, with the intervals between successive bats entering the leaf less than half the interval from unfamiliar groups (Fig. 3D). We also created simulated groups by combining the entry times of solitary bats finding and entering the leaf (Fig. 3D) to gain an insight into the size of the effect of familiarity relative to the benefits from grouping alone. We found that the time between successive entries in familiar groups was less than one quarter the time expected from simulated groups (Fig. 3D). This confirms that coordination in this collective task is substantially

greater when bats are familiar rather than unfamiliar, and that this effect is over-and-above the benefits of grouping alone.

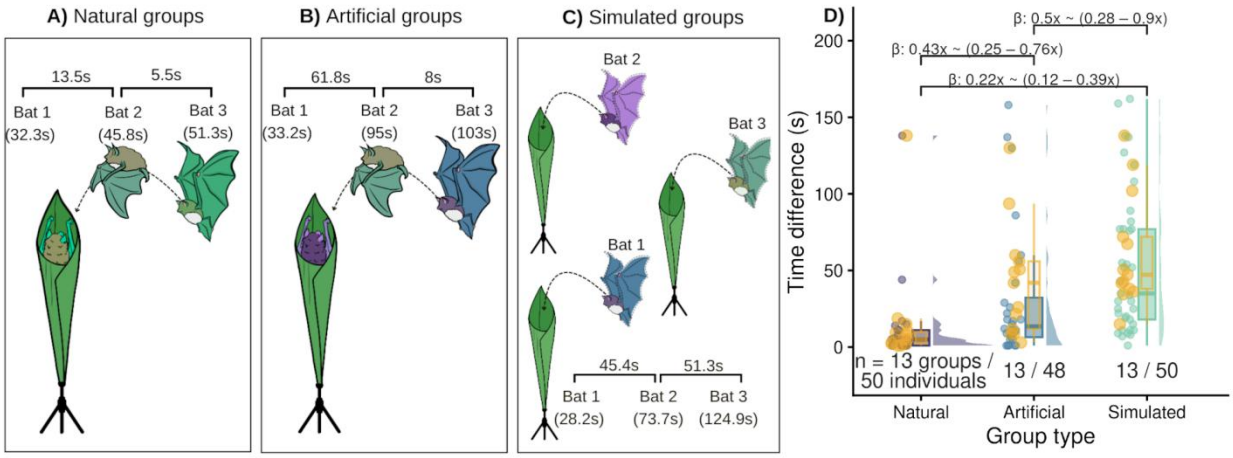

Fig. 3. Scheme of the experimental design to assess roost entry synchrony and its associated results. Synchronization was quantified as the time lapse between successive individuals when entering the roost. Artificial groups (**B**) were composed by individuals from different natural groups (**A**) while simulated groups (**C**) were digitally created by combining entry times from single individual trials. (**D**) Raincloud plots showing the time difference (s) between successive individuals entering a roost. Brackets in panels (A–C) indicate the mean entry times of the first three bats and the corresponding mean intervals between successive entries. Jittered points represent individual-level observations, with associated boxplots (median, interquartile range, and overall range) and half-violin plots shown for each group flight type. Yellow points indicate group-level mean time differences, with their distribution summarized by overlaid yellow boxplots. Brackets above boxplots denote statistically detectable differences between group types for the time between successive entries, with effect sizes and 95% uncertainty intervals expressed as fraction relative to the larger time span. Sample sizes for groups and individuals are shown below each boxplot. Full model structure and results are available in the supplementary materials.

Vocal coordination

Maintaining close spatial cohesion plays an important role in allowing bats to remain together while navigating the dense rainforest and finding roosts. Such precise coordination—in both space and time—relies on effective communication. Communication plays a central role in the performance of human teams (*13*, *30*, *31*) and has been linked to increased familiarity (*13*, *32*). However, complex environments—such as rainforests—introduce barriers to effective communication. Fish schools that experience reduced access to visual information, such as in turbid waters, increases cohesion (*33*) but make slower and less accurate decisions (*34*), while ants deposit (and rely more on) pheromones in low-light conditions (*35*, *36*). Birds, which have

access to both visual and vocal communication systems, are more likely to switch to making acoustic alarm calls when in longer than when in shorter grass (*37*). Spix's disc-winged bats have limited visual information both due to vegetation density and due to flying in the dark, but employ a dual call system for coordinating movements and locating roosts. Flying individuals constantly emit ultrasonic inquiry calls while searching for a roost, which are typically answered by response calls from individuals already inside the roost (*38*). However, a challenge for acoustic communication is that it can easily suffer from information masking or signal jamming (*39*). This 'cocktail party effect' (*40–42*) arises when many individuals are trying to vocalise simultaneously.

Mitigating the cocktail party effect requires coordinating the delivery of inquiry calls to minimize temporal overlap without compromising the overall vocal output to maintain sufficient information transmission. We recorded calls of solitary bats, natural (familiar) groups and artificial (unfamiliar) groups, applying individual call identification models in group settings to obtain data on call rates. We found distinct differences in how bats called in natural versus artificial groups. Both natural and artificial groups had lower call overlap than expected by chance (simulated groups created by combining solitary flights; Fig. 4A), indicating active avoidance of signal jamming at the group level. This seems to be achieved by adjusting call rates at the individual level, as call rates per bat during group flights were lower than those in solo flights (Fig. 4C). However, group-level call rates (number of all calls emitted by all bats of a group per minute) by natural groups were nearly twice as high as the call rates by artificial groups (and similar to those of solitary individuals, Fig. 4D), suggesting that familiar groups can produce a collective signal that maintains a higher level of acoustic information than unfamiliar groups. Higher vocal coordination in natural groups seems to be achieved by individual level adjustments rather than from a few individuals dominating calling activity as the proportion of calling individuals was similar between natural and artificial groups (Fig. 4B).

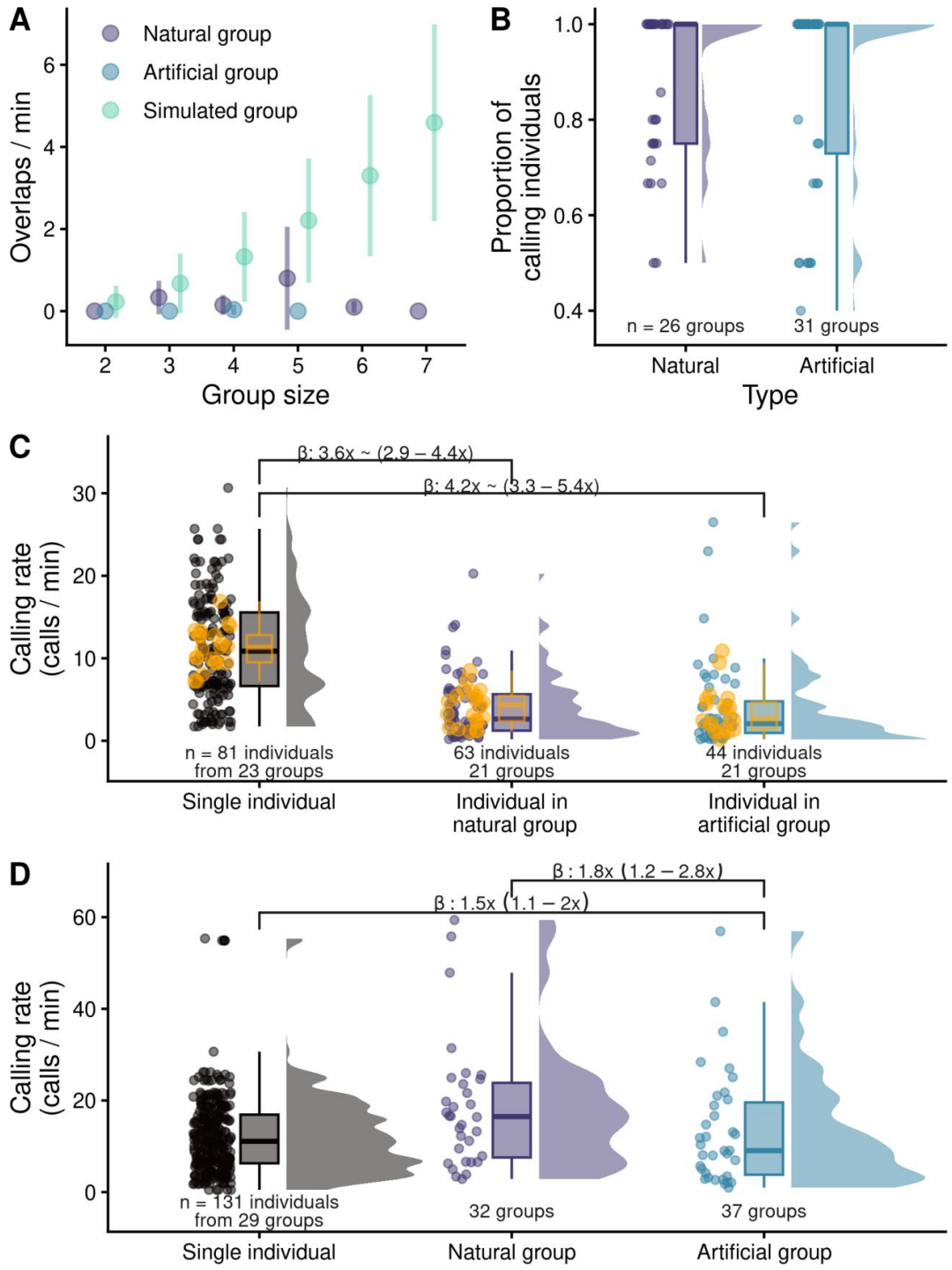

Fig. 4. Vocal coordination and familiarity. (**A**) Average number of inquiry call overlaps in group flights. Simulated values were obtained from 1000 iterations of a custom routine that combined single individual calling data from solo flights. (**B**) Proportion of individuals within a group that called at least once in a group flight. (**C**) Individual inquiry call rate for individuals in solo flights vs call rates of individuals in natural and artificial groups. (**D**) Individual inquiry call rate for single individuals in solo flights vs group-level call rates (total calls of all individuals in a group

per min) from natural and artificial groups. Data in (B), (C) and (D) is shown as raincloud plots of raw data points, boxplots (median, interquartile range and overall range) and half violin plots. Brackets above boxplots indicate statistically detectable differences with effect size and 95% uncertainty intervals as multiplicative differences in call rate. Sample sizes for groups and individuals are shown underneath boxplots. Yellow points show the mean call rate per group with their distribution represented by overlaid yellow boxplots. Full model structure and results are available in the supplementary materials.

The distinct differences in how natural and artificial groups performed at solving an ecologically relevant task highlights the important contribution of familiarity to collective behaviour by non-human animals. The benefits of familiarity appear to arise specifically through more effective communication, as previously found in human groups (*13*, *32*). In addition, unfamiliarity itself can impose costs: groups of strangers often function less efficiently because individuals must devote attention to evaluating and predicting the behaviour of unfamiliar partners, a cognitively demanding task that can detract from coordination (*43*, *44*). The importance of familiarity and its role in facilitating communication and coordination in various collective tasks that we have uncovered in this study is likely to have widespread importance among social animals. Maintaining cohesion, precisely coordinating actions, and making synchronous collective movements, underpins many ecological processes ranging from reproductive ecology (*43*, *45–48*) through to migration (*49–51*) that are frequently undertaken in familiar groups. Our findings therefore extend the concept of "team familiarity" beyond humans, showing it to be a general principle of collective life. By demonstrating that even small bat societies rely on familiarity to succeed in critical ecological challenges, this work highlights a common thread linking human collaboration with the collective strategies that shape animal societies across taxa.

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List of Supplementary Materials:

Methods and Results
Supplementary Text
Statistical analysis details and R code